```
import pandas as pd
df = pd.read_excel("Decode labs P1 - Cleaned Data set.xlsx")
df["Date"] = pd.to_datetime(df["Date"])
df.head()
```

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMeth
0	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit C
1	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	On
2	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit C
3	ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit C
4	ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	On

```
df[["Quantity","UnitPrice","TotalPrice"]].describe()
```

	Quantity	UnitPrice	TotalPrice
count	1200.000000	1200.000000	1200.000000
mean	2.945833	356.412750	1053.968300
std	1.407557	197.177146	819.856558
min	1.000000	11.390000	11.390000
25%	2.000000	186.062500	410.520000
50%	3.000000	364.210000	823.615000
75%	4.000000	521.570000	1578.475000
max	5.000000	699.930000	3456.400000

```
df["TotalPrice"].mean(), df["TotalPrice"].median()
```

```
(np.float64(1053.9683), 823.615)
```

```
q1 = df["TotalPrice"].quantile(0.25)
q3 = df["TotalPrice"].quantile(0.75)
iqr = q3 - q1
low, high = q1 - 1.5*iqr, q3 + 1.5*iqr
df[(df["TotalPrice"] < low) | (df["TotalPrice"] > high)]
```

	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentM
107	ORD200107	2023-03-27	C16775	Printer	5	670.75	848 Main St	G
326	ORD200326	2024-07-01	C65986	Laptop	5	670.48	273 Main St	G
328	ORD200328	2023-02-28	C18404	Tablet	5	674.04	546 Main St	
469	ORD200469	2023-11-26	C13877	Chair	5	676.98	893 Main St	
632	ORD200632	2023-05-02	C67260	Laptop	5	678.16	463 Main St	G
789	ORD200789	2023-08-17	C57276	Tablet	5	691.28	183 Main St	
1065	ORD201065	2023-10-30	C47778	Printer	5	666.80	488 Main St	Deb
1122	ORD201122	2023-06-07	C38840	Monitor	5	678.19	766 Main St	

```
df["Product"].value_counts()
df.groupby("Product")["TotalPrice"].sum().sort_values(ascending=False)
df.groupby(df["Date"].dt.to_period("M"))["TotalPrice"].sum()
```

Date	TotalPrice
2023-01	56685.75
2023-02	40117.66
2023-03	48609.37
2023-04	27751.71
2023-05	63836.84
2023-06	49500.19
2023-07	42820.66
2023-08	54352.14
2023-09	29526.67
2023-10	52607.85
2023-11	43079.67
2023-12	43754.73
2024-01	38528.08
2024-02	36909.57
2024-03	36030.90
2024-04	49613.14
2024-05	27909.11
2024-06	68068.54
2024-07	42963.98
2024-08	31991.07
2024-09	39794.98
2024-10	37226.97
2024-11	32413.76
2024-12	38785.77
2025-01	29099.40
2025-02	35317.55
2025-03	39200.66
2025-04	31821.20
2025-05	43396.64
2025-06	53047.40

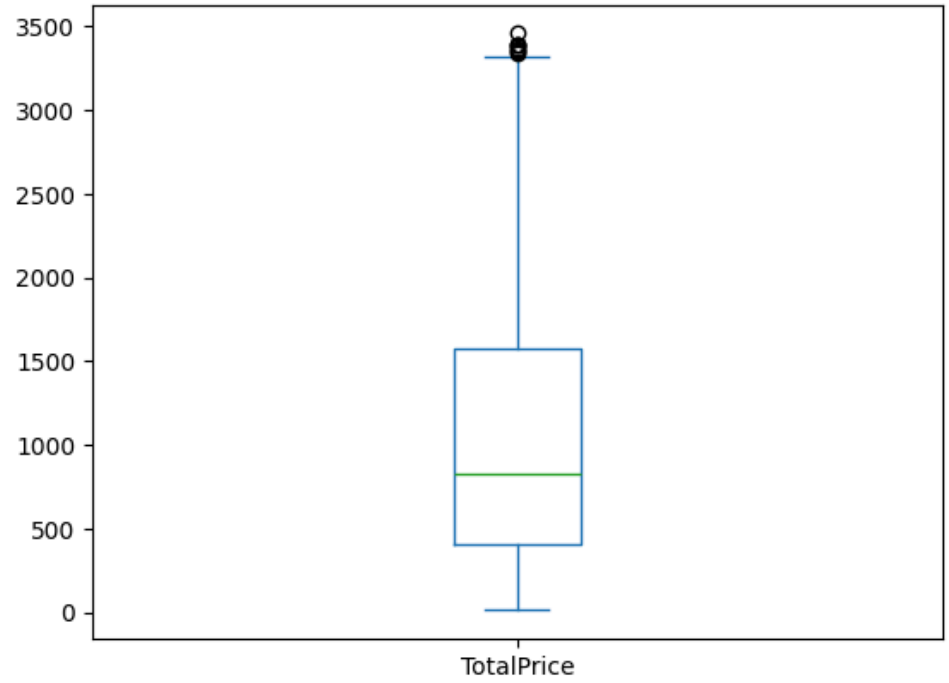
dtype: float64

```
df[["Quantity","UnitPrice","TotalPrice"]].corr()
```

	Quantity	UnitPrice	TotalPrice
Quantity	1.000000	0.014553	0.615251
UnitPrice	0.014553	1.000000	0.717081
TotalPrice	0.615251	0.717081	1.000000

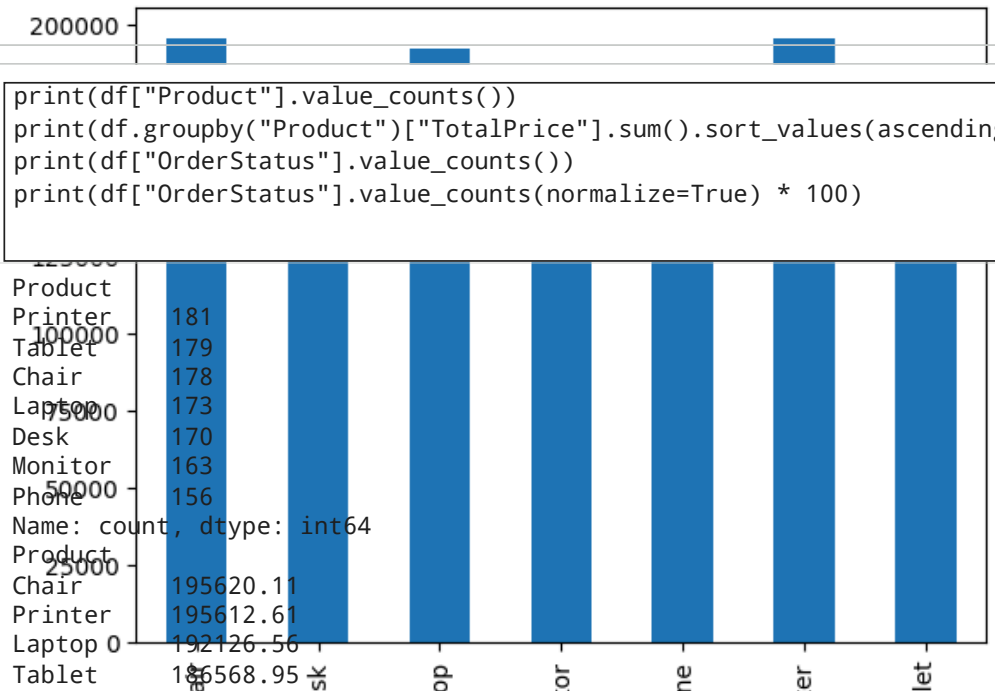
```
df["TotalPrice"].plot(kind="box")
```

```
<Axes: >
```




```
df.groupby("Product")["TotalPrice"].sum().plot(kind="bar")
```

```
<Axes: xlabel='Product'>
```




```
print(df["Product"].value_counts())
print(df.groupby("Product")["TotalPrice"].sum().sort_values(ascending=False))
print(df["OrderStatus"].value_counts())
print(df["OrderStatus"].value_counts(normalize=True) * 100)
```

```
Product
Printer    181
Tablet     179
Chair      178
Laptop     173
Desk       170
Monitor    163
Phone      156
Name: count, dtype: int64
Product
Chair      195620.11
Printer    195612.61
Laptop     192126.56
Tablet     186568.95
```